THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241 Introduction to Mathematical Analysis
Test 2 Solutions

1. (a) Convergent. Note that $-1 \leqslant \sin n \leqslant 1$ for any $n \in \mathbb{N}$. As $f(x) = \tan x$ is increasing on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, and $\tan(-1) = -\tan 1$, we have $|\tan(\sin n)| \leqslant \tan 1$ for any $n \in \mathbb{N}$. Thus, $\left|\frac{\tan(\sin n)}{n^3}\right| \leqslant \frac{\tan 1}{n^3}$. As the 3-series converges, so is $\sum_{n=1}^{\infty} \frac{\tan 1}{n^3}$. By the comparison test, the given series converges absolutely (hence converges).

   (b) Divergent. For any $n \geqslant 2$, we have $\frac{1}{\ln(2n^n+1)} > \frac{1}{\ln(3n^n)} > \frac{1}{\ln(n^{2n})} = \frac{1}{2n\ln n}$. Recall from Example 3.14 that the series $\sum_{n=2}^{\infty} \frac{1}{n\ln n}$ diverges. So the series $\sum_{n=2}^{\infty} \frac{1}{2n\ln n}$ diverges. By the comparison test, the given series diverges.

2. f is only continuous at -1.

   For any $a \in \mathbb{R}$, by the denseness of $\mathbb{Q}$, we can find a sequence $\{x_n\}$ of rational numbers converging to a with $x_n \neq a$ for all $n \in \mathbb{N}$. Then $\{f(x_n)\} = \{x_n^2 - 1\}$ converges to $a^2 - 1$. Similarly, by the denseness of $\mathbb{R} - \mathbb{Q}$, we can find a sequence $\{y_n\}$ of irrational numbers converging to a with $y_n \neq a$ for all $n \in \mathbb{N}$. Then $\{f(y_n)\} = \left\{-\frac{(y_n+1)(3y_n+1)}{y_n}\right\}$ either converges to $-\frac{(a+1)(3a+1)}{a}$ (when $a \neq 0$) or diverges (when $a = 0$ since it is unbounded). By the sequential criterion, f is continuous at a only if $a^2 - 1 = -\frac{(a+1)(3a+1)}{a}$, i.e. $(a+1)(a^2+2a+1) = 0$. The only solution is $a = -1$. This shows f is discontinuous at any point $a \neq -1$.

   When $a = -1$, we use the definition to prove that f is continuous at -1. For any $\varepsilon > 0$, we choose $\delta = \min\left\{\frac{1}{2}, \frac{\varepsilon}{3}\right\}$. For any $x \in \mathbb{R}$ satisfying $|x - (-1)| < \delta$, we can show that $|f(x) - f(-1)| < \varepsilon$ as follows.

   - If $x \in \mathbb{Q}$, then

     $$|f(x) - f(-1)| = |(x^2 - 1) - 0| = |x+1||(x+1) - 2| \leqslant |x+1|(|x+1| + 2) < \delta(\delta + 2) < 3\delta \leqslant \varepsilon.$$

   - If $x \notin \mathbb{Q}$, since $-\frac{3}{2} \leqslant x \leqslant -\frac{1}{2}$, we have

     $$|f(x) - f(-1)| = \left|-\frac{(x+1)(3x+1)}{x} - 0\right| = |x+1|\left|3 + \frac{1}{x}\right| < \delta \cdot 1 < \varepsilon.$$

   By definition, f is continuous at -1, and we are done.

3. For any $\varepsilon > 0$, we choose $\delta = \dfrac{\varepsilon}{8} > 0$. For any $x, y \in \mathbb{R}$ with $|x - y| < \delta$, we have

$$|g(x) - g(y)| = \left|\frac{2x+1}{x^2+2} - \frac{2y+1}{y^2+2}\right| = \left|\frac{2xy^2 + 4x + y^2 - 2x^2y - 4y - x^2}{(x^2+2)(y^2+2)}\right| = |y - x|\left|\frac{2xy + y + x - 4}{(x^2+2)(y^2+2)}\right|$$
$$< \delta\left(\frac{2|x||y|}{(x^2+2)(y^2+2)} + \frac{|y|}{(x^2+2)(y^2+2)} + \frac{|x|}{(x^2+2)(y^2+2)} + \frac{4}{(x^2+2)(y^2+2)}\right).$$

Note that $\dfrac{|x|}{x^2+2} < 1$ since this is equivalent to $|x|^2 - |x| + 2 > 0$, which holds as the discriminant $(-1)^2 - 4(1)(2)$ is negative. Also, it is obvious that $\dfrac{1}{x^2+2} < 1$. The same bounds apply if we replace x by y. Therefore, we have

$$|g(x) - g(y)| < \delta(2 + 1 + 1 + 4) = 8\delta = \varepsilon.$$

This shows g is uniformly continuous.

4. By the extreme value theorem, f attains its minimum value m. Let $S = \{x \in [a, b] : f(x) = m\}$. Note that S is a nonempty bounded set. Thus, $\sup S$ exists. We claim that $\sup S \in S$.

Suppose on the contrary that $\sup S \notin S$. Then $\sup S$ is a limit point of S, and there exists a sequence $\{x_n\}$ in S converging to $\sup S$ (for example, choose $x_n \in S \cap \left(\sup S - \dfrac{1}{n}, \sup S\right)$ for each n). As f is continuous, by the sequential criterion, $f(\sup S) = \lim\limits_{n\to\infty} f(x_n) = \lim\limits_{n\to\infty} m = m$. This yields the contradiction $\sup S \in S$.

Next, we claim that $\sup S = b$. If $\sup S < b$, then we use the condition with $x = \sup S$ and any $\delta > 0$. We can find $y \in [a, b] \cap (\sup S, \sup S + \delta)$ such that $f(y) = f(\sup S) = m$. This yields $y \in S$ but $y > \sup S$, contradicting $\sup S$ is an upper bound for S. Therefore, we must have $\sup S = b$, and hence $f(b) = m$

Using the same argument, if M is the maximum value of f, we can show that $f(b) = M$ (in fact, the same argument applies if we replace m by any real number in the range of f). This forces $m = M$. As the minimum and the maximum values coincide, the function must be a constant function.

5. (a) True. Note that $\sqrt{\dfrac{a_n^2 + 1}{2a_n}} > \dfrac{\sqrt{a_n}}{\sqrt{2}}$. By the comparison test, $\sum\limits_{n=1}^{\infty} \sqrt{\dfrac{a_n^2 + 1}{2a_n}}$ diverges.

(b) True. The 'if part' is trivial because $\{b_n\}$ is a subsequence of itself. For the 'only if' part, let the sequences of partial sums of $\sum\limits_{n=1}^{\infty} b_n$ and $\sum\limits_{k=1}^{\infty} b_{n_k}$ be $\{s_n\}$ and $\{t_k\}$ respectively. Then $t_k \leqslant s_{n_k}$. As $\{s_n\}$ is bounded, so is $\{t_k\}$, and hence the series $\sum\limits_{k=1}^{\infty} b_{n_k}$ converges.

(c) False. Let $f(x) = \begin{cases} 0 & \text{if } x = n^{-1} \text{ for some } n \in \mathbb{N}, \\ 1 & \text{if } x \neq n^{-1} \text{ for all } n \in \mathbb{N} \end{cases}$. Then the series converges since every term is 0, but $\lim\limits_{x\to 0} f(x)$ does not exist because $\lim\limits_{n\to\infty} f\left(\dfrac{1}{n}\right) = 0$ and $\lim\limits_{n\to\infty} f\left(\dfrac{\sqrt{2}}{n}\right) = 1$, but both sequences $\left\{\dfrac{1}{n}\right\}$ and $\left\{\dfrac{\sqrt{2}}{n}\right\}$ converge to 0.